

A Radial-Twist Counterexample to a Coarse-Sum Coordinate Problem

Automated Math Discovery Smoke Test

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Source and Open Question

This packet concerns Problem 5.7 from:

Bruno de Mendonca Braga, *Asymptotic structure and coarse Lipschitz geometry of Banach spaces*, arXiv:1604.08661; Studia Mathematica 237 (2017), 71–97.

The source location is page 20 of the arXiv PDF. In the parsed source, the problem appears at lines 980–984 of `source.tex`.

$$d(cC)^{-\delta} - Ck^{-\delta} \leq 2K Ck^{\delta/42-\delta/12} C^{-\delta} + K Ck^{\delta/42-\delta} + 3\delta$$

for all $k \in \mathbb{N}$, and all $\delta \in (0, 1)$. As $q_2 > p > 1$, by letting $k \rightarrow \infty$ and $\delta \rightarrow 0$, we get a contradiction. $\square$

If $T = (T_1, T_2) : X \rightarrow Y_1 \oplus Y_2$ is a linear isomorphic embedding, then either $T_1 : X \rightarrow Y_1$ or $T_2 : X \rightarrow Y_2$ is not strictly singular, i.e., $T_i : X_0 \rightarrow Y_i$ is a linear isomorphic embedding, for some infinite dimensional subspace $X_0 \subset X$, and some $i \in \{1, 2\}$. Is there an analog of this result for coarse Lipschitz embeddings? Precisely, we ask the following.

Problem 5.7. Let X, Y_1 and Y_2 be Banach spaces and consider a coarse Lipschitz embedding $f = (f_1, f_2) : X \rightarrow Y_1 \oplus Y_2$. Is there an infinite dimensional subspace $X_0 \subset X$ such that either $f_1 : X_0 \rightarrow Y_1$ or $f_2 : X_0 \rightarrow Y_2$ is a coarse Lipschitz embedding?

We can now prove Theorem 1.6, which will be essential in the proof of Theorem 1.7.

Proof of Theorem 1.6. Say $m \in \{1, \dots, n-1\}$ is such that $p \in (p_m, p_{m+1})$ (the other cases have analogous proofs). Then $(T^{p_{m+1}} \oplus \dots \oplus T^{p_n})_\epsilon$ is reflexive (see

Problem 5.7 asks whether, for every coarse Lipschitz embedding

$$f = (f_1, f_2) : X \rightarrow Y_1 \oplus Y_2,$$

there is an infinite-dimensional subspace $X_0 \subset X$ such that either $f_1|_{X_0} : X_0 \rightarrow Y_1$ or $f_2|_{X_0} : X_0 \rightarrow Y_2$ is a coarse Lipschitz embedding.

Answer

The answer is negative. We construct a bi-Lipschitz embedding

$$F = (F_1, F_2) : \ell_2 \rightarrow \ell_2 \oplus_2 \ell_2$$

such that neither coordinate is a coarse embedding on any infinite-dimensional subspace of ℓ_2 .

Construction

Let $H = \ell_2$ be a real Hilbert space. Define

$$d_4(t) = \text{dist}(t, 4\mathbb{Z}), \quad \varphi(t) = \frac{\pi}{4} \min\{d_4(t), 2\} \quad (t \geq 0),$$

and

$$\theta(r) = \varphi(\log(1+r)) \quad (r \geq 0).$$

Then φ is $\pi/4$ -Lipschitz, takes values in $[0, \pi/2]$, and

$$\theta(e^{4k} - 1) = 0, \quad \theta(e^{4k+2} - 1) = \frac{\pi}{2}$$

for all $k \in \mathbb{N} \cup \{0\}$.

For $x \in H$, set

$$F(x) = (\cos(\theta(\|x\|))x, \sin(\theta(\|x\|))x) \in H \oplus_2 H.$$

Write $F = (F_1, F_2)$ for its two coordinate maps.

The Full Map is Bi-Lipschitz

Theorem 1. *For all $x, y \in H$,*

$$\frac{1}{\sqrt{2}} \|x - y\| \leq \|F(x) - F(y)\|_{H \oplus_2 H} \leq \left(1 + \frac{\pi}{4}\right) \|x - y\|.$$

In particular, F is a coarse Lipschitz embedding.

Proof. For $t \in [0, \pi/2]$, define the isometry

$$J_t : H \rightarrow H \oplus_2 H, \quad J_t z = (\cos t z, \sin t z).$$

Then $F(x) = J_{\theta(\|x\|)}x$.

Let $r = \|x\|$ and $s = \|y\|$. If both x and y are nonzero, write $u = x/r$, $v = y/s$, $a = \langle u, v \rangle$, and $c = \cos(\theta(r) - \theta(s))$. Since θ takes values in $[0, \pi/2]$, we have $c \in [0, 1]$. A direct calculation gives

$$\|F(x) - F(y)\|^2 = r^2 + s^2 - 2rsca,$$

whereas

$$\|x - y\|^2 = r^2 + s^2 - 2rsa.$$

If $a \geq 0$, then $ca \leq a$, so

$$\|F(x) - F(y)\| \geq \|x - y\|.$$

If $a < 0$, then

$$\|F(x) - F(y)\|^2 \geq r^2 + s^2 \geq \frac{1}{2} \|x - y\|^2.$$

The same lower estimate is immediate if one of x, y is zero. Hence

$$\|F(x) - F(y)\| \geq 2^{-1/2} \|x - y\|.$$

For the upper estimate, assume without loss of generality that $r \geq s$. Then

$$\begin{aligned}\|F(x) - F(y)\| &\leq \|J_{\theta(r)}(x - y)\| + \|(J_{\theta(r)} - J_{\theta(s)})y\| \\ &\leq \|x - y\| + s|\theta(r) - \theta(s)|.\end{aligned}$$

Since φ is $\pi/4$ -Lipschitz,

$$|\theta(r) - \theta(s)| \leq \frac{\pi}{4} \log \frac{1+r}{1+s}.$$

Using $\log(1+u) \leq u$ with $u = (r-s)/(1+s)$, we get

$$s \log \frac{1+r}{1+s} \leq \frac{s}{1+s}(r-s) \leq r-s \leq \|x-y\|.$$

Therefore

$$\|F(x) - F(y)\| \leq \left(1 + \frac{\pi}{4}\right) \|x - y\|.$$

□

The Coordinate Maps Fail on Every Infinite-Dimensional Subspace

Theorem 2. *Let $X_0 \subset H$ be any nonzero linear subspace. Then neither $F_1|_{X_0}$ nor $F_2|_{X_0}$ is a coarse embedding. Consequently, the conclusion asked for in Problem 5.7 fails for every infinite-dimensional subspace $X_0 \subset H$.*

Proof. Choose $u \in X_0$ with $\|u\| = 1$.

For F_1 , set

$$R_k = e^{4k+2} - 1, \quad x_k = R_k u.$$

Then $\theta(R_k) = \pi/2$, so

$$F_1(x_k) = F_1(-x_k) = 0.$$

However,

$$\|x_k - (-x_k)\| = 2R_k \rightarrow \infty.$$

Thus the compression modulus of $F_1|_{X_0}$ is zero at arbitrarily large scales, so $F_1|_{X_0}$ is not a coarse embedding.

For F_2 , set

$$S_k = e^{4k} - 1, \quad y_k = S_k u.$$

Then $\theta(S_k) = 0$, so

$$F_2(y_k) = F_2(-y_k) = 0,$$

while

$$\|y_k - (-y_k)\| = 2S_k \rightarrow \infty.$$

Therefore $F_2|_{X_0}$ is not a coarse embedding either.

□

Conclusion

Taking $X = Y_1 = Y_2 = \ell_2$ and using the map $F = (F_1, F_2)$ above gives a coarse Lipschitz embedding into the two-coordinate direct sum, but no coordinate map is a coarse Lipschitz embedding on any infinite-dimensional subspace. This gives a negative answer to Problem 5.7.

Remark 1. *The proof was written for the Hilbert two-sum $Y_1 \oplus_2 Y_2$. If the direct sum in the problem is read with the ℓ_1 norm, the same example still works: on $Y_1 \times Y_2$ the ℓ_1 and ℓ_2 norms are equivalent, so the full map remains a coarse Lipschitz embedding, and the coordinate-collapse argument is unchanged.*

Verification Notes

The verification script in the `code` folder samples the same construction in finite-dimensional Hilbert space. It checks that the sampled distance ratios lie within the proved constants

$$2^{-1/2} \quad \text{and} \quad 1 + \pi/4,$$

and verifies the exact coordinate collapses at the radii $e^{4k+2} - 1$ and $e^{4k} - 1$.

Novelty Check

I searched for the exact text of Problem 5.7 and for later papers using the phrases “coarse Lipschitz embedding”, “ $Y_1 \oplus Y_2$ ”, and “strictly singular”. The search found the source paper and related coarse-Lipschitz geometry papers, but no later paper explicitly answering this problem. This packet should still be treated as a candidate result until a human checks the surrounding literature more thoroughly.